

Q. 1 Chemistry

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- A N
- B ON
- C CO
- D c Cl
- C

Q. 3 Chemistry

Assertion : HClO_4 is stronger acidic than HClO_3 . Reason : Oxidation state of Cl in HClO_4 is + VII and in HClO_3 + V.

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- A If both assertion and reason are true and the reason is the correct explanation of the assertion.
 - B If both assertion and reason are true but reason is not the correct explanation of the assertion.
 - C If assertion is true but reason is false.
 - D If the assertion and reason both are false.
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- Reason : Oxidation state of Cl in HClO_4 is + VII and in HClO_3 + V.
- If both assertion and reason are true but reason is not the correct explanation of the assertion.
- If the assertion and reason both are false.

Q. 5 Chemistry

In which reaction there is a change in valency

- A $2\text{NO}_2 \rightarrow \text{N}_2\text{O}_4$
- B $2\text{NO}_2 + \text{H}_2\text{O} \rightarrow \text{HNO}_2 + \text{HNO}_3$
- C $\text{NH}_4\text{OH} \rightarrow \text{NH}_4^+ + \text{OH}^-$
- D $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$

Q. 7 Chemistry

How many moles of $\text{K}_2\text{Cr}_2\text{O}_7$ can be reduced by 1 mole of Sn^{2+}

- A 1/3
- B 1/6
- C 2/3
- D 1

Q. 9 Chemistry

The compound $\text{YBa}_2\text{Cu}_3\text{O}_7$ which shows superconductivity has copper in oxidation state Assume that the rare earth element Yttrium is in its usual + 3 oxidation state

- A 3/7
- B 7/3
- C 3
- D 7

Q. 11 Chemistry

Which one of the following reactions does not involve either oxidation or reduction

- A $\text{VO}_{+2} \rightarrow \text{V}_2\text{O}_3$
- B $\text{Na} \rightarrow \text{Na}^+$
- C $\text{CrO}_{2-4} \rightarrow \text{Cr}_2\text{O}_{2-7}$
- D $\text{Zn}^{2+} \rightarrow \text{Zn}$

Q. 13 Chemistry

Assertion : Fluorine exists only in - 1 oxidation state. Reason : Fluorine has $2s^2 2p^5$ configuration.

- A If both assertion and reason are true and the reason is the correct explanation of the assertion.
- B If both assertion and reason are true but reason is not the correct explanation of the assertion.
- C If assertion is true but reason is false.
- D If the assertion and reason both are false.

Q. 15 Chemistry

The oxidation number of sulphur in S_8 , S_2F_2 , H_2S respectively, are

- A 0, + 1 and - 2
- B + 2, + 1 and - 2
- C 0, + 1 and + 2
- D - 2, + 1 and - 2

Q. 17 Chemistry

Oxidation state of oxygen atom in potassium superoxide is

- A 0
B - 1
C -12
D - 2

Q. 19 Chemistry

In the reaction $\text{I}_2 + 2\text{S}_2\text{O}_{\text{---}3} \rightarrow 2\text{I}^- + \text{S}_4\text{O}_{\text{---}6}$ equivalent weight of iodine will be equal to

- A 1/2 of molecular weight
B Molecular weight
C 1/4 of molecular weight
D None

Q. 21 Chemistry

Match List I with List II and select the correct answer using the codes given below the lists

	List I (Compound)	List II (Oxidation state of N)	
3	(C) NH_3	(3) + 4	(D) N_2O_5
		(4) + 1	Codes :

- A A B C D2 3 4 1
B A B C D3 1 2 4
C A B C D3 4 2 1
D A B C D2 3 1 4

Q. 23 Chemistry

The equivalent weight of phosphoric acid (H_3PO_4) in the reaction $\text{NaOH} + \text{H}_3\text{PO}_4 \rightarrow \text{NaH}_2\text{PO}_4 + \text{H}_2\text{O}$ is

- A 25
B 49
C 59
D 98

Q. 25 Chemistry

The oxidation number of nitrogen in NH_2OH is

- A + 1
B - 1
C - 3
D - 2

Q. 27 Chemistry

The strongest reducing agent is

- A HNO_2
B H_2S
C H_2SO_3
D SnCl_2

Q. 29 Chemistry

Which one of the following reactions is not an example of redox reaction

- A $\text{Cl}_2 + 2\text{H}_2\text{O} + \text{SO}_2 \rightarrow 4\text{H}^+ + \text{SO}_4^{4-} + 2\text{Cl}^-$
B $\text{Cu}^{++} + \text{Zn} \rightarrow \text{Zn}^{++} + \text{Cu}$
C $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$
D $\text{HCl} + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{Cl}^-$

Q. 31 Chemistry

In which of the following reactions H_2O_2 is a reducing agent

- A $2\text{FeCl}_2 + 2\text{HCl} + \text{H}_2\text{O}_2 \rightarrow 2\text{FeCl}_3 + 2\text{H}_2\text{O}$
B $\text{Cl}_2 + \text{H}_2\text{O}_2 \rightarrow 2\text{HCl} + \text{O}_2$
C $2\text{HI} + \text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{I}_2$
D $\text{H}_2\text{SO}_3 + \text{H}_2\text{O}_2 \rightarrow \text{H}_2\text{SO}_4 + \text{H}_2\text{O}$

Q. 33 Chemistry

The oxidation number of nitrogen in NO_{-3} is

- A - 1
B + 2
C + 3
D + 5

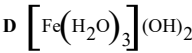
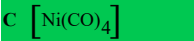
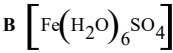
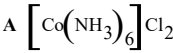
Q. 35 Chemistry

The oxidation number of Fe and S in iron pyrites are

- A 4, - 2
B 2, - 1
C 3, - 1.5
D 3, - 1

Q. 37 Chemistry

In which of the following compounds transition metal is in oxidation state zero



Q. 39 Chemistry

Which of the following statements is correct

A Hydrogen has oxidation number -1 and $+1$

B Hydrogen has same electronegativity as halogens

C Hydrogen will not be liberated at anode

D Hydrogen has same ionization potential as alkali metals

Q. 41 Chemistry

Oxidation number of cobalt in $\text{K}\left[\text{Co}\left(\text{CO}\right)_4 \right]$ is

A $+1$

B $+3$

C -1

D -3

Q. 43 Chemistry

Oxidation number of Fe in $\text{K}_3\left[\text{Fe}\left(\text{CN}\right)_6 \right]$ is

A $+2$

B $+3$

C $+1$

D $+4$

Q. 45 Chemistry

The number of electrons involved in the reduction of $\text{Cr}_2\text{O}_{2-7}$ in acidic solution to Cr^{3+} is

A 0

B 2

C 3

D 5